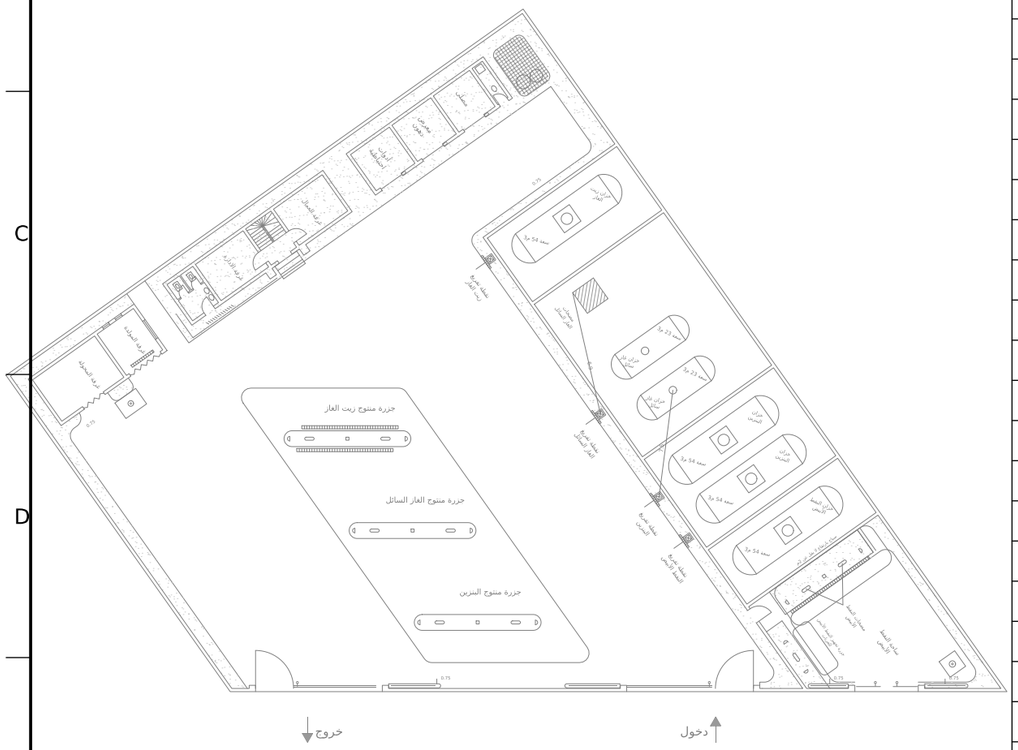
FUEL FILLING STATION

محطة تعبئة وقود - القطعة المرقمة (1 / 3453) مقاطعة (43) يارمجة الشرقية
قضاء الموصل / محافظة نينوى - للمواطنين (هادي حسين جمعة - احمد عثمان عيد)

ELECTRICAL - CIVIL DEFENCE - SANITARY - STRUCTURAL

DESIGN DRAWINGS, ISSUE 00 (STAGE 0: GENERAL SHEETS)

OWNERS: HADI HUSSEIN JUMAA - AHMED OTHMAN EID



SITE PLAN (REFERENCE - SEE FUEL_STATION_SITE_PLAN.dxf)
PLOT AREA 2,415.25 m2 - STREET FRONT 61.28 m
ABOVE-GROUND STORAGE 262 m3 IN 6 TANKS - 9 DISPENSERS

DRAWING REGISTER

No.	DRAWING	TITLE	STATUS
01	G-001	COVER SHEET - DRAWING REGISTER	ISSUED FOR REVIEW
02	G-002	LEGEND - SYMBOLS AND LINE TYPES	ISSUED FOR REVIEW
03	G-003	DESIGN BASIS - CODES, DATA AND GENERAL NOTES	ISSUED FOR REVIEW
04	F-001	SAFETY DISTANCES - SEPARATION PLAN AND COMPLIANCE TABLE	PLANNED - STAGE 1
05	F-002	FIRE WATER NETWORK - HYDRANTS, HOSE REELS, LPG SPRAY COOLING	PLANNED - STAGE 1
06	F-003	PORTABLE EXTINGUISHERS - LAYOUT AND QUANTITY SCHEDULE	PLANNED - STAGE 1
07	F-004	FIRE ALARM AND ESD - DETECTION, CALL POINTS, SHUT-DOWN	PLANNED - STAGE 1
08	F-005	EVACUATION PLAN - EXITS, ROUTES, TANKER CIRCULATION	PLANNED - STAGE 1
09	F-006	FIRE WATER SCHEMATIC - PUMP SET AND RISER DIAGRAM	PLANNED - STAGE 1
10	F-007	BUND COMPARTMENTS - CAPACITY CHECK, DRAIN VALVES, SUMPS	PLANNED - STAGE 1
11	F-008	CIVIL DEFENCE DETAILS - SIGNS, CABINETS, MOUNTING	PLANNED - STAGE 1
12	E-001	POWER SUPPLY ARRANGEMENT - TRANSFORMER, GENERATOR, MDB, ATS	PLANNED - STAGE 2
13	E-002	HAZARDOUS AREAS - ZONE 0/1/2 CLASSIFICATION AND SECTIONS	PLANNED - STAGE 2
14	E-003	EXTERNAL POWER - DISPENSER AND TANK CABLES, TRENCHES	PLANNED - STAGE 2
15	E-004	EXTERNAL LIGHTING - YARD AND BOUNDARY (CANOPY IN STAGE 6)	PLANNED - STAGE 2
16	E-005	BUILDINGS LIGHTING - LAYOUT AND CIRCUITS	PLANNED - STAGE 2
17	E-006	BUILDINGS POWER - SOCKETS, A/C AND CIRCUITS	PLANNED - STAGE 2
18	E-007	SINGLE LINE DIAGRAM - MDB, SMDBs, ATS AND ESD	PLANNED - STAGE 2
19	E-008	LOAD SCHEDULES - MAXIMUM DEMAND AND PANEL SCHEDULES	PLANNED - STAGE 2
20	E-009	EARTHING AND BONDING - EARTH NETWORK AND TANKER STATIC EARTH	PLANNED - STAGE 2
21	E-010	LIGHTNING PROTECTION - BUILDINGS AND TANKS (CANOPY IN STAGE 6)	PLANNED - STAGE 2
22	P-001	WATER SUPPLY - MAINS, GROUND TANK, PUMP, DISTRIBUTION	PLANNED - STAGE 3
23	P-002	FOUL DRAINAGE - MANHOLES, SEPTIC TANK, SOAKAWAY	PLANNED - STAGE 3
24	P-003	OILY AND SURFACE WATER - DRAINS AND OIL SEPARATOR	PLANNED - STAGE 3
25	P-004	TOILETS - ENLARGED PLANS AND ISOMETRICS	PLANNED - STAGE 3
26	P-005	SANITARY DETAILS - SEPTIC TANK, SEPARATOR, MANHOLES	PLANNED - STAGE 3
27	S-001	STRUCTURAL NOTES - MATERIALS, LOADS, SEISMIC DATA	PLANNED - STAGE 4
28	S-002	BUILDINGS FOUNDATIONS - FOOTINGS AND TIE BEAMS	PLANNED - STAGE 4
29	S-003	BUILDINGS FRAME - COLUMNS, BEAMS, ROOF SLAB, STAIR	PLANNED - STAGE 4
30	S-004	TANK SADDLES - SADDLES AND FOUNDATIONS	PLANNED - STAGE 4
31	S-005	BUND WALLS - LIQUID-RETAINING RC WALLS	PLANNED - STAGE 4
32	S-006	BURIED STRUCTURES - SEPTIC TANK, SEPARATOR, FIRE TANK	PLANNED - STAGE 4
33	S-007	SITE STRUCTURES - BOUNDARY WALL, PIERS, PLINTHS	PLANNED - STAGE 4
34	S-008	CANOPY FOUNDATIONS - AFTER ARCHITECTURAL CANOPY DESIGN	HOLD - CANOPY DESIGN
35	S-009	CANOPY STEEL STRUCTURE - AFTER ARCHITECTURAL CANOPY DESIGN	HOLD - CANOPY DESIGN

STAGE 6 (S-008, S-009 AND THE CANOPY PARTS OF E-004, E-008, E-010, P-003) STARTS AFTER THE ARCHITECTURAL 3D ELEVATIONS AND THE CANOPY FORM ARE APPROVED.

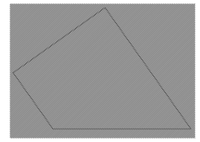
GENERAL

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GENERAL NOTES

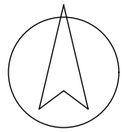
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KEY PLAN



THIS SHEET

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TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
COVER SHEET
DRAWING REGISTER

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
G-001

SHEET
01
OF 3

CIVIL DEFENCE

SYMBOL	CODE	DESCRIPTION	الوصف
P	FE-P	PORTABLE EXTINGUISHER, DRY POWDER 12 kg	مطفأة بودرة جافة 12 كغم
C	FE-C	PORTABLE EXTINGUISHER, CO2 6 kg	مطفأة ثاني اوكسيد الكاربون 6 كغم
F	FE-F	PORTABLE EXTINGUISHER, AFFF FOAM 9 L	مطفأة رغوة 9 لتر
50	FE-W	WHEELED EXTINGUISHER, DRY POWDER 50 kg	مطفأة بودرة جافة بعجلات 50 كغم
	FH	FIRE HYDRANT (2 x 65 mm OUTLETS)	حنفية حريق
HR	HR	FIRE HOSE REEL 25 mm x 30 m	بكرة خرطوم حريق
FACP	FACP	FIRE ALARM CONTROL PANEL	لوحة سيطرة انذار الحريق
S	SD	SMOKE DETECTOR	كاشف دخان
H	HD	HEAT DETECTOR	كاشف حرارة
MCP	MCP	MANUAL CALL POINT	نقطة انذار يدوية
B	SNDR	ALARM SOUNDER / FLASHER	جرس وضوء انذار
ESD	ESD	EMERGENCY SHUT-DOWN PUSH BUTTON	زر ايقاف الطوارئ
EXIT	EXIT	EMERGENCY EXIT SIGN (ILLUMINATED)	علامة مخرج طوارئ
AP	AP	ASSEMBLY POINT	نقطة تجمع
FP	FP	FIRE PUMP SET	مضخة حريق
	WS	WATER SPRAY NOZZLE (LPG TANK COOLING)	مرش ماء لتبريد خزان الغاز

STRUCTURAL

SYMBOL	CODE	DESCRIPTION	الوصف
	GRID	GRID LINE BUBBLE	محور انشائي
	COL	RC COLUMN	عمود خرساني
	FTG	ISOLATED FOOTING	اساس منفرد
	SEC	SECTION MARK	علامة مقطع

ELECTRICAL

SYMBOL	CODE	DESCRIPTION	الوصف
	TRF	TRANSFORMER 11/0.4 kV	محولة 11/0.4 ك.ف
G	GEN	STANDBY DIESEL GENERATOR	مولدة احتياط
ATS	ATS	AUTOMATIC TRANSFER SWITCH	مفتاح تحويل اوتوماتيكي
	MDB	MAIN / SUB DISTRIBUTION BOARD	لوحة توزيع رئيسية / فرعية
	LP	LED PANEL 600x600, 36 W	وحدة انارة LED
	LEX	Ex-d LED LUMINAIRE (HAZARDOUS AREA)	وحدة انارة مقاومة للانفجار
EX	POLE	YARD LIGHTING POLE, LED FLOODLIGHT	عمود انارة ساحة
	EM	EMERGENCY LIGHT, 3 h BATTERY	انارة طوارئ
	SO	SWITCHED SOCKET OUTLET 13 A	مأخذ قدرة 13 امبير
	SW	LIGHTING SWITCH	مفتاح انارة
	JB	JUNCTION BOX	صندوق توصيل
	EP	EARTH PIT WITH TEST LINK	بئر تأريض
AT	AT	LIGHTNING AIR TERMINAL	مانعة صواعق

SANITARY

SYMBOL	CODE	DESCRIPTION	الوصف
	MH	MANHOLE / INSPECTION CHAMBER	منهول / غرفة تفتيش
	GT	GULLY TRAP	مصيدة مياه
	FD	FLOOR DRAIN	بالوعة ارضية
CO	CO	CLEAN-OUT	فتحة تنظيف
M	WM	WATER METER	مقياس ماء
	GV	GATE VALVE	صمام بوابي
	NRV	NON-RETURN VALVE	صمام عدم رجوع
	PU	BOOSTER PUMP	مضخة رفع

LINE TYPES

LINE	LAYER	DESCRIPTION	الوصف
	F-PIPE	FIRE WATER MAIN (HDPE PN16 BURIED)	خط ماء الحريق
	E-POWR	BURIED POWER CABLE IN TRENCH / DUCT	كابل قدرة مدفون
	E-GRND	BARE COPPER EARTH CONDUCTOR 70 mm2	موصل تأريض نحاسي
	E-HAZD	HAZARDOUS AREA BOUNDARY (ZONE 1 / 2)	حدود المنطقة الخطرة
	P-WATR	COLD WATER SUPPLY (PPR)	خط ماء بارد
	P-SANR	FOUL DRAIN (uPVC)	خط صرف صحي
	P-STRM	OILY / SURFACE WATER DRAIN	خط تصريف مياه سطحية / زيتية

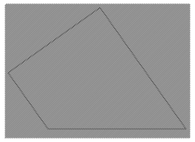
GENERAL

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GENERAL NOTES

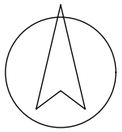
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KEY PLAN



THIS SHEET

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TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
LEGEND
SYMBOLS AND LINE TYPES

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE	DATE
NTS	18/10/2026
PHASE: DESIGN DEVELOPMENT	STATUS: FOR REVIEW

DRAWING No.
G-002

SHEET
02
OF 3

PROJECT DATA

SITE

PLOT AREA 2,415.25 m2 (RIGHT TRAPEZOID 50 x 30.59 / 66.02)

STREET FRONT 61.28 m, ONE EXIT 10 m, ONE ENTRY 10 m, YARD GATE 5 m

PRODUCTS AND STORAGE (ALL TANKS ABOVE GROUND IN RC BUNDS)

GAS OIL 1 TANK(S) 54 m3

LPG 2 TANK(S) 46 m3

GASOLINE 2 TANK(S) 108 m3

KEROSENE 1 TANK(S) 54 m3

TOTAL 262 m3 - BUND COMPARTMENTS C1 (GAS OIL), C2 (LPG), C3 (PETROL, KEROSENE)

DISPENSING

I-1 GAS OIL 2 DISPENSER(S)

I-2 LPG 2 DISPENSER(S)

I-3 GASOLINE 2 DISPENSER(S)

I-4 KEROSENE (YARD) 2 DISPENSER(S)

I-5 KEROSENE (HAND CARTS) 1 DISPENSER(S)

UTILITIES

POWER: 11 kV SUPPLY, DEDICATED 11/0.4 kV TRANSFORMER IN THE TRANSFORMER ROOM, STANDBY DIESEL GENERATOR WITH AUTOMATIC TRANSFER SWITCH.

WATER: MUNICIPAL MAINS + GROUND STORAGE TANK + BOOSTER PUMP.

DRAINAGE: NO PUBLIC SEWER - SEPTIC TANK AND SOAKAWAY; OILY WATER THROUGH AN OIL SEPARATOR BEFORE DISPOSAL.

DESIGN ASSUMPTIONS (TO BE CONFIRMED)

ALLOWABLE SOIL BEARING 100 kN/m2 PENDING A SOIL INVESTIGATION.

BUILDINGS: REINFORCED CONCRETE FRAME AND SOLID ROOF SLAB.

CANOPY: FORM AND STRUCTURE PENDING ARCHITECTURAL DESIGN (STAGE 6).

NORTH POINT NOT SHOWN ON THE SOURCE PLAN - TO BE CONFIRMED ON SITE.

GENERAL NOTES

1. THIS PACKAGE IS BASED ON THE PRELIMINARY SITE PLAN (FUEL_STATION_SITE_PLAN.dxf) REDRAWN FROM ITS WRITTEN DIMENSIONS. ANY CHANGE TO THE SITE PLAN IS TO BE REFLECTED IN ALL DISCIPLINES.
2. ALL SAFETY DISTANCES ARE CHECKED ON F-001; ANY SHORTFALL IS REPORTED, NOT HIDDEN.
3. ALL ELECTRICAL EQUIPMENT INSIDE ZONE 1 / ZONE 2 SHALL BE CERTIFIED FOR THE ZONE (Ex d / Ex e, IIA T3 MINIMUM, IIB FOR LPG AREAS).
4. ALL CALCULATED VALUES SHOWN ON THE DRAWINGS COME FROM THE DESIGN REPORT (calcs.py).
5. THE DRAWINGS ARE FOR AUTHORITY SUBMISSION AND REVIEW AND REQUIRE THE SIGNATURE OF A REGISTERED ENGINEER (IRAQI ENGINEERS UNION).
6. CANOPY-DEPENDENT ITEMS ARE ON HOLD UNTIL THE ARCHITECTURAL CANOPY DESIGN IS APPROVED (STAGE 6).

IRAQI CODES AND AUTHORITY REQUIREMENTS

IRAQI BUILDING CODES (IRAQI BUILDING COUNCIL - MINISTRY OF CONSTRUCTION AND HOUSING):

مدونة التأسيسات الكهربائية - الطبعة الاولى 2016

المواصفات الفنية للأعمال الكهربائية 2015

مدونة التأسيسات المائية في المباني

المواصفات الفنية للأعمال الصحية

مدونات الاحمال والخرسانة المسلحة والزلازل العراقية

CIVIL DEFENCE DIRECTORATE REQUIREMENTS (مديرية الدفاع المدني).

OIL PRODUCTS DISTRIBUTION COMPANY (MINISTRY OF OIL):

INSTRUCTIONS FOR PRIVATE FUEL FILLING STATIONS - LATEST EDITION TO BE OBTAINED; CLAUSES MARKED (OPDC) ARE TO BE VERIFIED AGAINST IT.

INTERNATIONAL STANDARDS

NFPA 30 / NFPA 30A FLAMMABLE LIQUIDS / MOTOR FUEL DISPENSING

NFPA 58 LIQUEFIED PETROLEUM GAS CODE

NFPA 10 / 15 / 72 EXTINGUISHERS / WATER SPRAY / FIRE ALARM

NFPA 70 ART. 514 MOTOR FUEL DISPENSING FACILITIES

IEC 60079-10-1 HAZARDOUS AREA CLASSIFICATION

EI MODEL CODE PART 15 AREA CLASSIFICATION (PETROLEUM)

IEC 60364 / IEC 62305 LV INSTALLATIONS / LIGHTNING PROTECTION

ACI 318 / ACI 350 STRUCTURAL CONCRETE / LIQUID-RETAINING

IPC INTERNATIONAL PLUMBING CODE (REFERENCE)

INFORMATION REQUIRED FROM THE AUTHORITIES / CLIENT

1. OPDC INSTRUCTIONS (LATEST EDITION) - SEPARATION DISTANCES, TANK AND DISPENSER REQUIREMENTS.
2. CIVIL DEFENCE DIRECTORATE - FIRE WATER STORAGE AND HYDRANT REQUIREMENTS FOR THE STATION CATEGORY.
3. ELECTRICITY DIRECTORATE - 11 kV CONNECTION POINT AND TRANSFORMER RATING.
4. MUNICIPAL WATER CONNECTION POINT AND PRESSURE.
5. SOIL INVESTIGATION REPORT (BEARING CAPACITY, GYPSUM CONTENT, WATER TABLE).
6. TRUE NORTH AND SITE LEVELS (TOPOGRAPHIC SURVEY).
7. ARCHITECTURAL CANOPY FORM, HEIGHT AND ELEVATIONS.

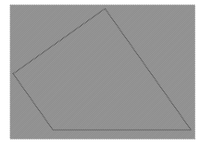
GENERAL

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GENERAL NOTES

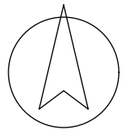
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KEY PLAN



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REVISIONS

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00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
DESIGN BASIS
CODES, DATA AND GENERAL NOTES

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

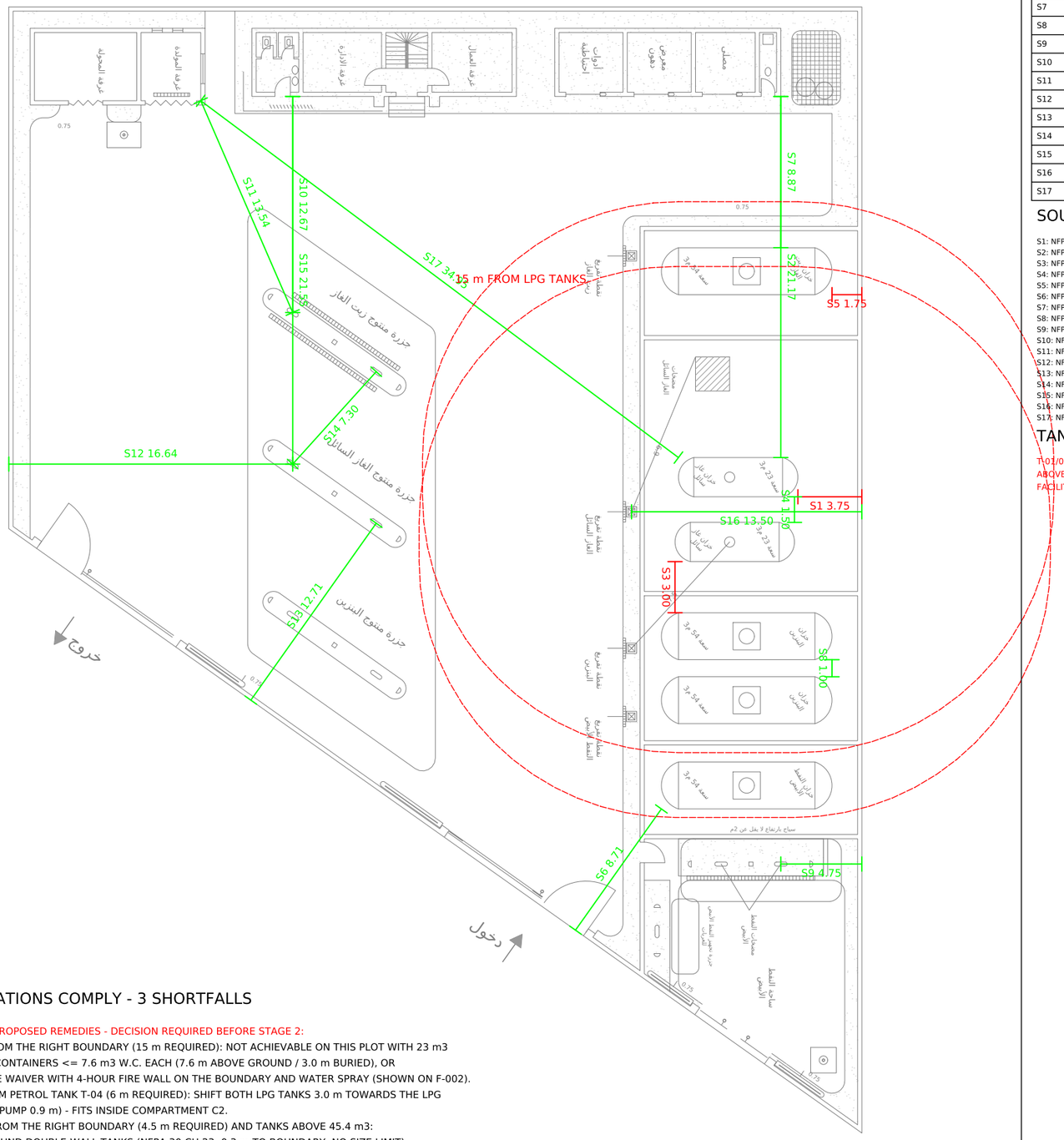
DATE
18/10/2026

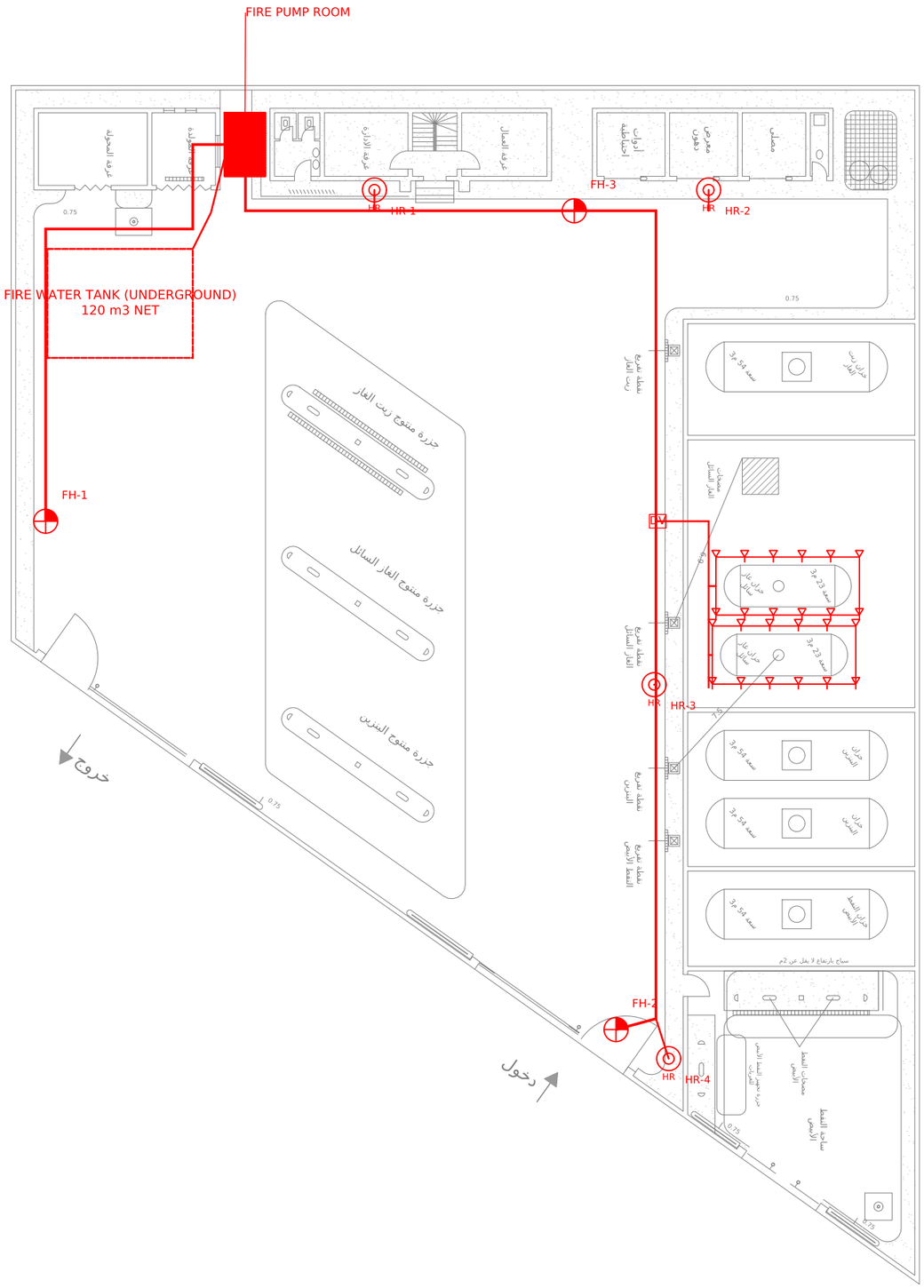
PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
G-003

SHEET
03
OF 3


15 m FROM LPG TANKS
S1 3.75
S2 21.17
S3 3.00
S4 1.50
S5 1.75
S6 8.71
S7 8.87
S8 11.00
S9 4.75
S10 12.67
S11 13.54
S12 16.64
S13 12.71
S14 7.30
S15 21.55
S16 13.50
S17 34.33
0.75
0.75
0.75



FIRE WATER DEMAND

LPG TANKS SURFACE (2 TANKS)	95.1 m2
WATER SPRAY 10.2 L/min/m2	970 L/min
HYDRANT HOSE STREAMS (2 x 475)	950 L/min
DESIGN FLOW	1920 L/min
DURATION	60 min
REQUIRED VOLUME	115.2 m3
FIRE TANK (NET)	120 m3

FIRE PUMPS (NFPA 20):
- MAIN ELECTRIC 2000 L/min @ 70 m
- STANDBY DIESEL 2000 L/min @ 70 m
- JOCKEY 50 L/min @ 80 m

SPECIFICATION

1. FIRE MAIN: HDPE PE100 PN16, 110 mm, BURIED 0.9 m MIN.; DUCTILE IRON ABOVE GROUND.
2. HYDRANTS: PILLAR TYPE 2 x 65 mm INSTANTANEOUS OUTLETS (IRAQI CIVIL DEFENCE COUPLING), ISOLATING VALVE ON EACH BRANCH.
3. HOSE REELS: 25 mm x 30 m, IN CABINETS.
4. LPG TANK COOLING: AUTOMATIC DELUGE VALVE (DV) OUTSIDE COMPARTMENT C2, OPEN NOZZLES ON A RING 0.45 m CLEAR OF EACH TANK, 10.2 L/min/m2 OVER THE WHOLE TANK SURFACE (NFPA 15).
5. DV OPENED BY HEAT/FLAME DETECTION OVER THE LPG TANKS AND BY MANUAL RELEASE AT THE DV.
6. FIRE TANK: RC, WATERPROOFED, TRAFFIC-RATED ROOF, FILLED FROM THE MAINS (P-001), WITH A CIVIL DEFENCE INLET BREECHING AT THE FRONT.
7. SEE F-006 FOR THE PUMP SET AND RISER DIAGRAM.

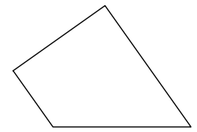
CIVIL DEFENCE

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GENERAL NOTES

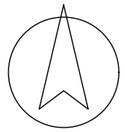
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KEY PLAN



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REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE WATER NETWORK

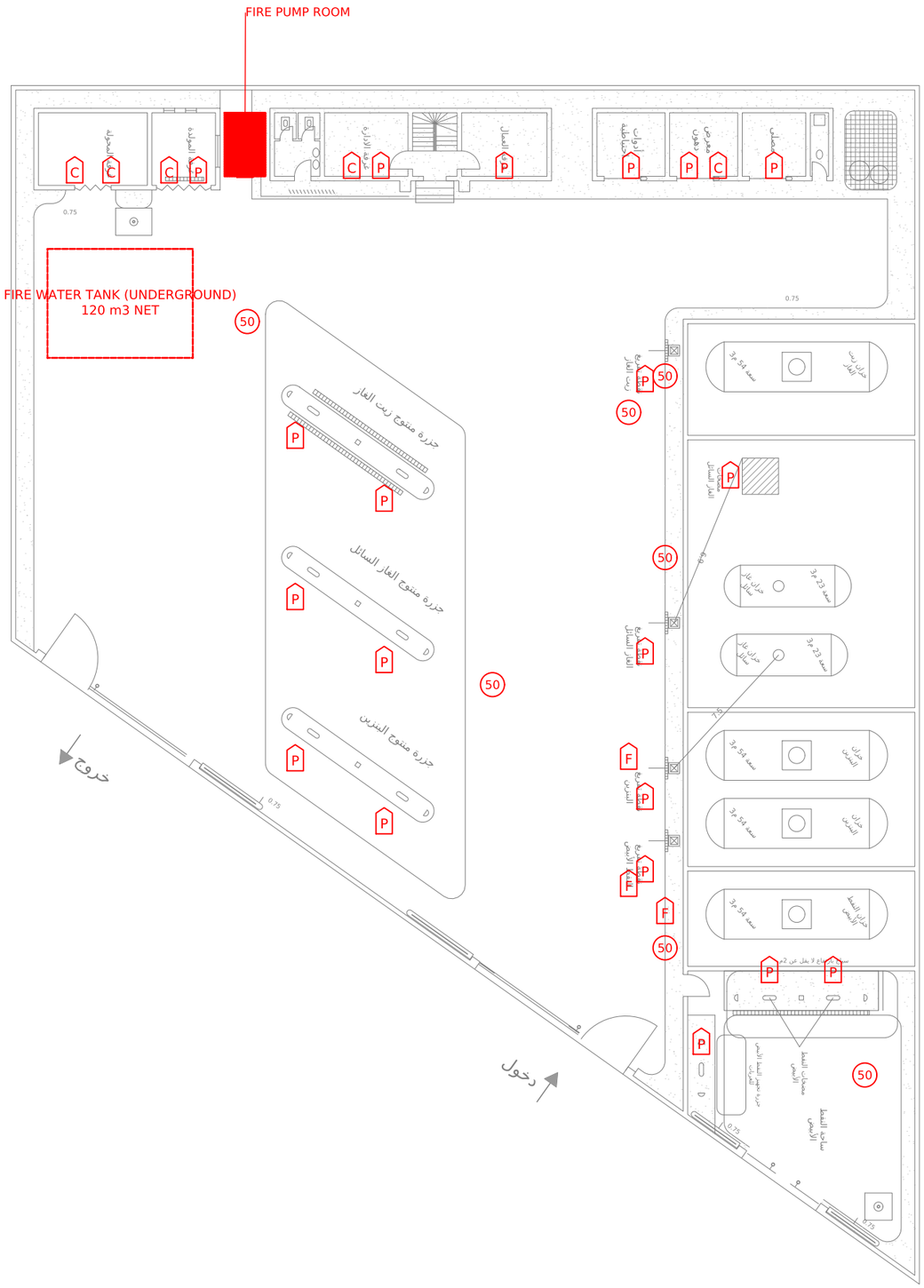
HYDRANTS, HOSE REELS, LPG SPRAY COOLING

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE	DATE
1:300	18/10/2026
PHASE: DESIGN DEVELOPMENT	STATUS: FOR REVIEW

DRAWING No.
F-002

SHEET
02
OF 8



LOCATION	TYPE	QTY
ISLAND I-1 (GAS OIL)	FE-P	2
ISLAND I-2 (LPG)	FE-P	2
ISLAND I-3 (GASOLINE)	FE-P	2
ISLAND I-4 (KEROSENE)	FE-P	2
ISLAND I-5 (KEROSENE (CARTS))	FE-P	1
ISLANDS AREA (CANOPY)	FE-W	2
KEROSENE YARD	FE-W	1
UNLOADING POINT - GAS OIL	FE-P	1
UNLOADING POINT - LPG	FE-P	1
UNLOADING POINT - GASOLINE	FE-P	1
UNLOADING POINT - KEROSENE	FE-P	1
UNLOADING POINTS	FE-W	1
UNLOADING POINTS	FE-F	2
TANK COMPARTMENT C1	FE-W	1
TANK COMPARTMENT C2 (LPG)	FE-W	1
TANK COMPARTMENT C3	FE-W	1
TANK COMPARTMENT C3	FE-F	1
LPG PUMP	FE-P	1
TRANSFORMER ROOM	FE-C	2
GENERATOR ROOM	FE-C	1
GENERATOR ROOM	FE-P	1
ADMINISTRATION (MDB)	FE-C	1
ADMINISTRATION	FE-P	1
WORKERS ROOM	FE-P	1
SPARE PARTS SHOP	FE-P	1
PAINTS SHOWROOM	FE-P	1
PAINTS SHOWROOM	FE-C	1
PRAYER ROOM	FE-P	1
FIRE PUMP ROOM	FE-P	1

TOTALS

FE-C PORTABLE EXTINGUISHER, CO2 6 kg: 5
FE-F PORTABLE EXTINGUISHER, AFFF FOAM 9 L: 3
FE-P PORTABLE EXTINGUISHER, DRY POWDER 12: 21
FE-W WHEELED EXTINGUISHER, DRY POWDER 50 : 7

NOTES

1. MOUNT PORTABLE UNITS WITH THE TOP <= 1.5 m ABOVE FLOOR, IN CABINETS OUTDOORS.
2. MAX. TRAVEL DISTANCE TO A UNIT 15 m (CLASS B).
3. CO2 ONLY IN ELECTRICAL ROOMS; NO WATER TYPE NEAR ELECTRICAL EQUIPMENT.
4. INSPECT MONTHLY, SERVICE YEARLY (NFPA 10).

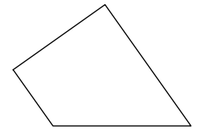
CIVIL DEFENCE

الدفاع المدني

GENERAL NOTES

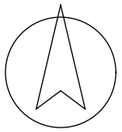
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KEY PLAN



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REVISIONS

REV	DATE	DESCRIPTION	BY
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PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

PORTABLE EXTINGUISHERS

LAYOUT AND QUANTITY SCHEDULE

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

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DATE
18/10/2026

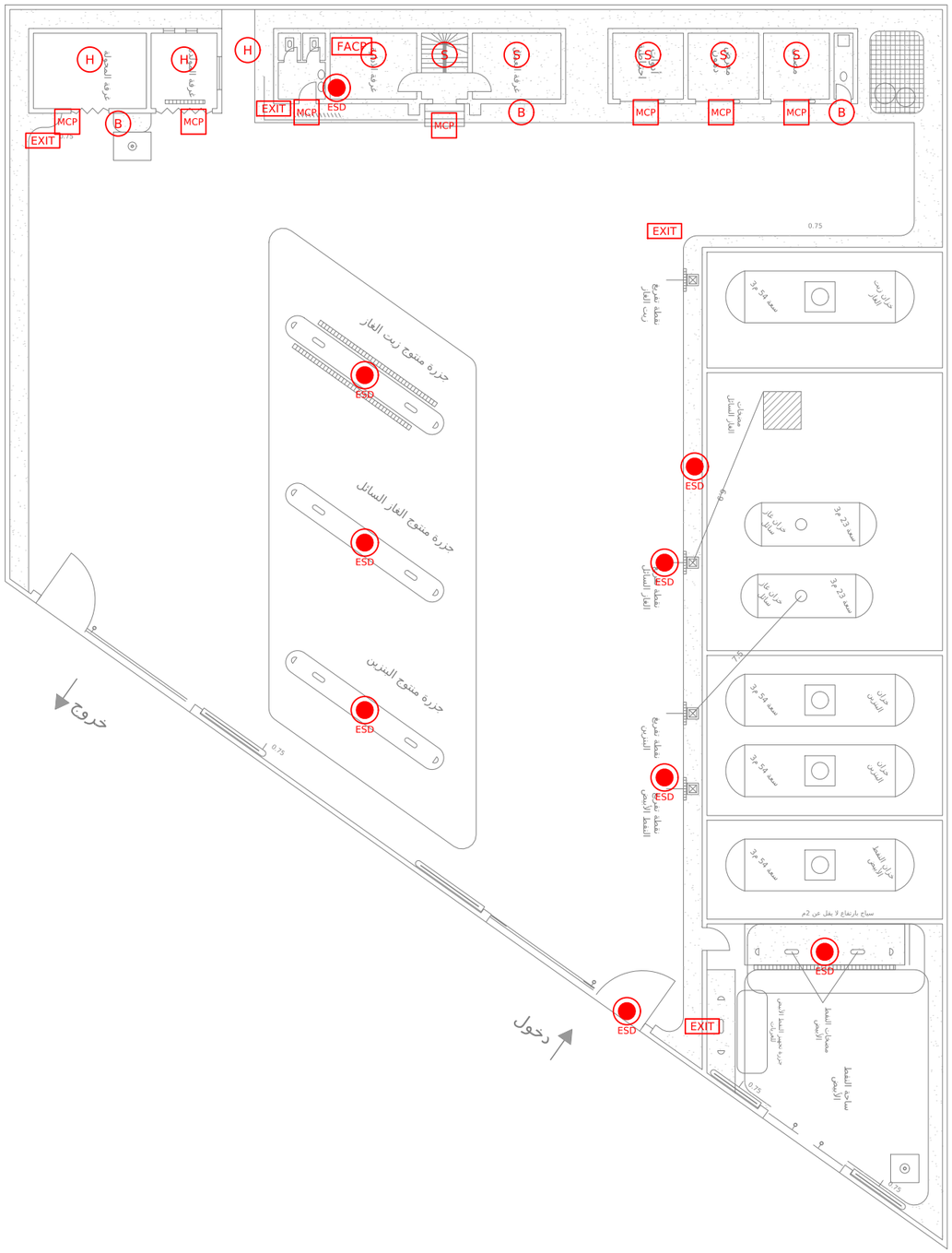
PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
F-003

SHEET
03
OF 8

FACP IN ADMINISTRATION



0 3 6 9 12 15m

SCALE 1:300 (A3)

FIRE ALARM

CONVENTIONAL 4-ZONE PANEL (NFPA 72 / EN 54):
Z1 BUILDING 1 (TRANSFORMER, GENERATOR)
Z2 BUILDING 2 (WC, ADMIN, STAIR, WORKERS)
Z3 BUILDING 3 (SHOPS, PRAYER, WC)
Z4 FIRE PUMP ROOM + LPG DELUGE STATUS
CABLE: FIRE-RESISTANT 2 x 1.5 mm2 (PH120).
STANDBY BATTERIES 24 h + 30 min ALARM.

EMERGENCY SHUT-DOWN (ESD)

9 RED MUSHROOM PUSH BUTTONS (STAY-PUT):
AT EVERY ISLAND, THE UNLOADING POINTS, THE
LPG COMPARTMENT, THE OFFICE AND THE GATE.
ANY BUTTON TRIPS: ALL DISPENSERS, SUBMERSIBLE
AND TRANSFER PUMPS, LPG PUMP AND THE CANOPY
POWER (NFPA 30A 6.7). FIRE PUMPS, FIRE ALARM
AND EMERGENCY LIGHTING STAY ON.
RESET ONLY AT THE PANEL IN THE OFFICE.
BUTTONS 6 - 30 m FROM DISPENSERS, IN VIEW.

SAFETY SIGNS (ARABIC + ENGLISH)

NO SMOKING / NO NAKED FLAME - AT EVERY ISLAND,
THE TANK AREA AND THE UNLOADING POINTS.
STOP ENGINE - SWITCH OFF MOBILE PHONES.
LPG - FLAMMABLE GAS (TANK COMPARTMENT C2).
EMERGENCY SHUT-DOWN - AT EVERY ESD BUTTON.
FIRE EXIT / ASSEMBLY POINT - SEE F-005.

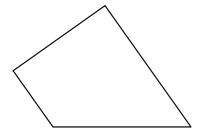
CIVIL DEFENCE

الدفاع المدني

GENERAL NOTES

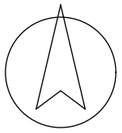
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PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE ALARM AND ESD

DETECTION, CALL POINTS, SHUT-DOWN

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
1:300

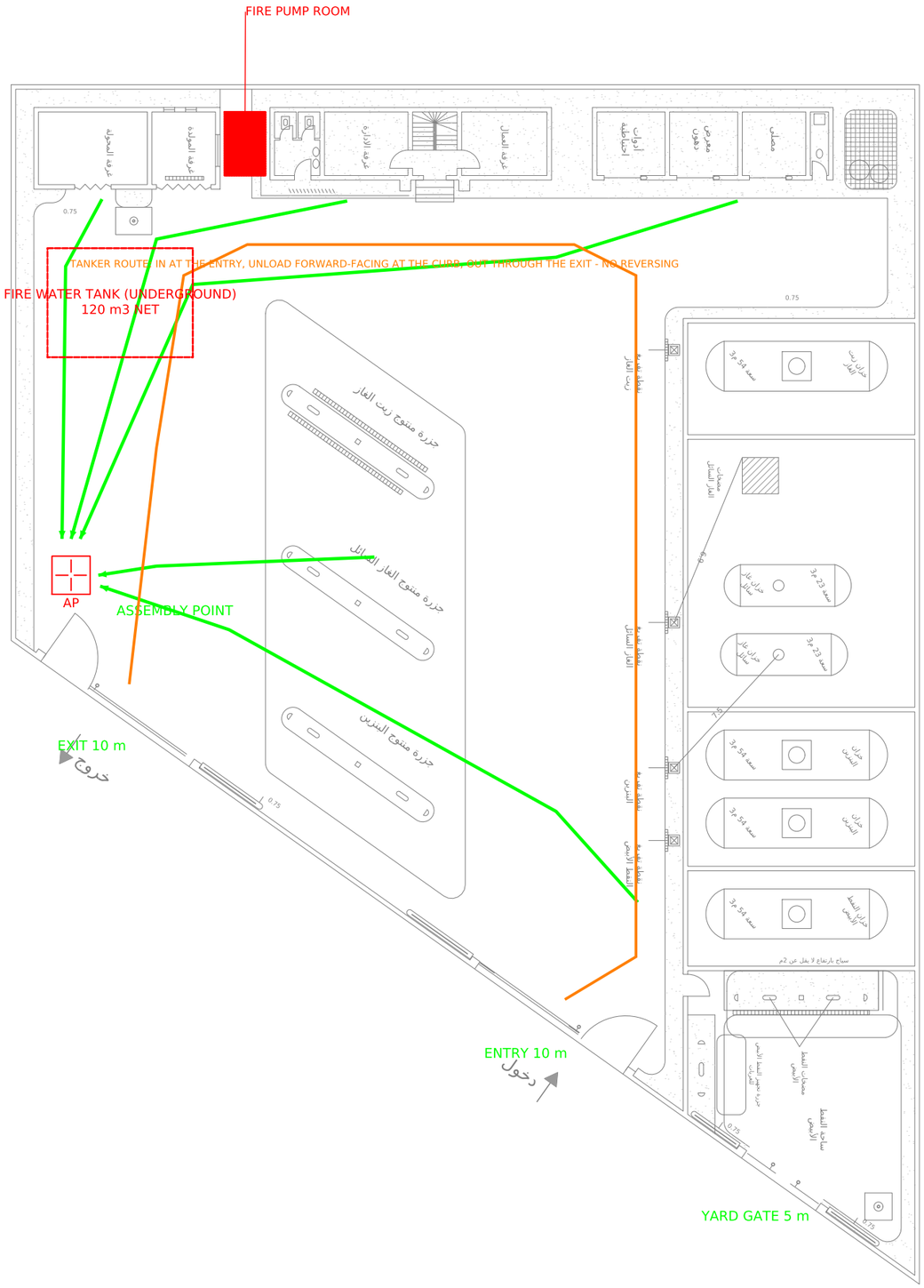
DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
F-004

SHEET
04
OF 8



SCALE 1:300 (A3)

EVACUATION

1. ON ALARM: STOP ALL DISPENSING (ESD), STOP UNLOADING, CLOSE TANKER VALVES.
2. STAFF AND CUSTOMERS WALK TO THE ASSEMBLY POINT AT THE FRONT-LEFT CORNER, AWAY FROM THE TANKS (> 30 m), AS SHOWN.
3. GATES STAY OPEN FOR CIVIL DEFENCE VEHICLES; CLEAR WIDTH 10 m AT ENTRY AND EXIT.
4. A TANKER UNLOADS ONLY WHEN THE STATION ATTENDANT, EXTINGUISHERS AND THE STATIC EARTH CLAMP ARE IN PLACE (E-009).
5. ROUTES ARE PAVED, LIT (E-004) AND KEPT FREE OF PARKED VEHICLES.

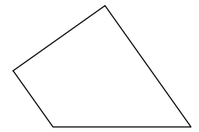
CIVIL DEFENCE

الدفاع المدني

GENERAL NOTES

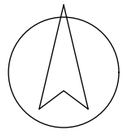
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REVISIONS

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PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

EVACUATION PLAN

EXITS, ROUTES, TANKER CIRCULATION

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
1:300

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

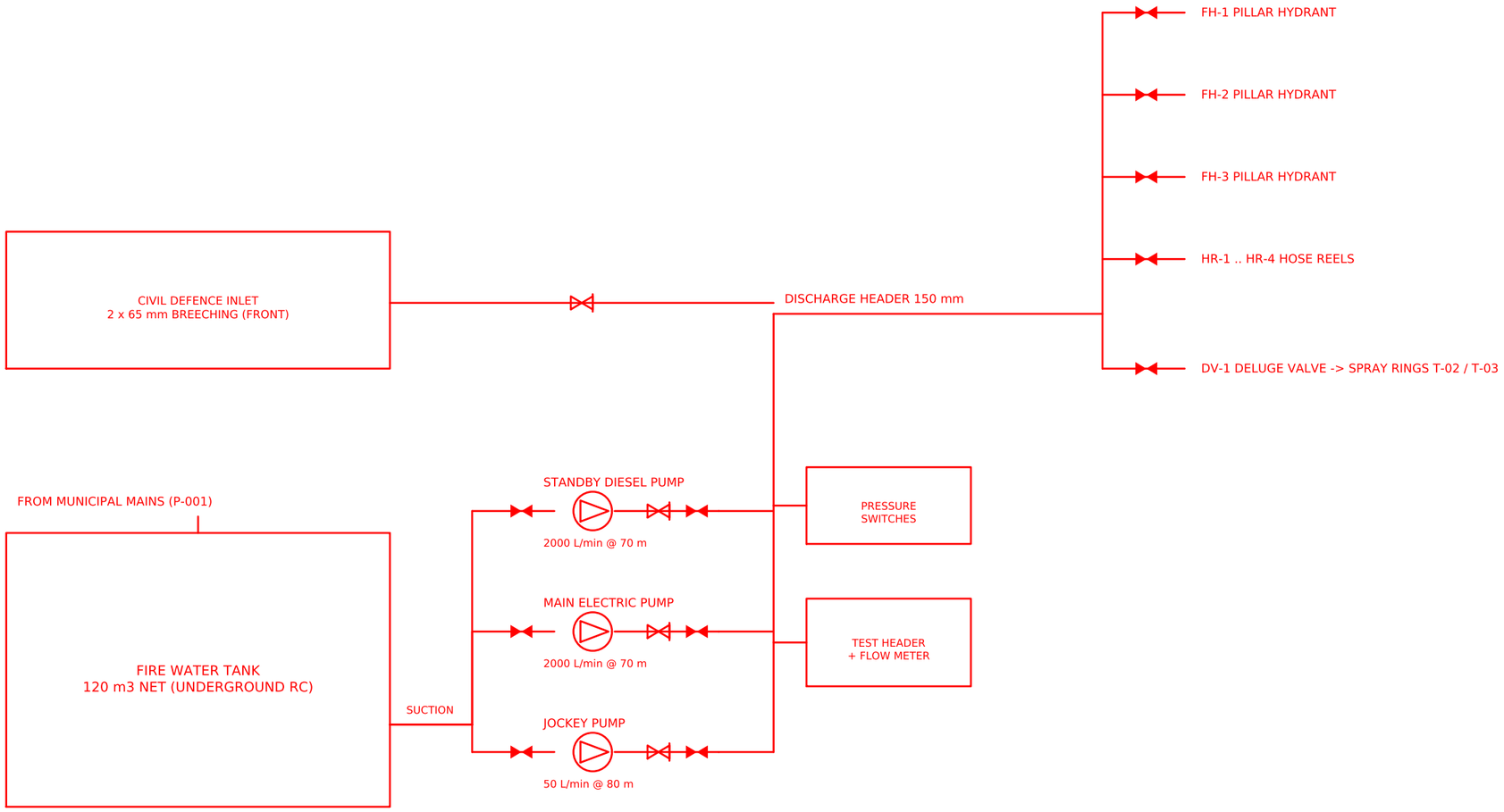
STATUS
FOR REVIEW

DRAWING No.
F-005

SHEET
05
OF 8

FIRE WATER SYSTEM - SCHEMATIC (NOT TO SCALE)

1. PUMP SET TO NFPA 20: JOCKEY STARTS AT 7.0 bar, MAIN AT 6.0 bar, DIESEL AT 5.5 bar; MAIN AND DIESEL STOP MANUALLY ONLY.
2. DIESEL PUMP WITH ITS OWN 8 h FUEL TANK AND TWO BATTERY SETS; EXHAUST LED OUTSIDE.
3. MAIN PUMP FED FROM THE MDB AHEAD OF THE MAIN BREAKER AND FROM THE GENERATOR (E-007).
4. DESIGN FLOW 1920 L/min FOR 60 min = 115.2 m3; TANK 120 m3 NET.
5. ALL VALVES LOCKED OPEN; PUMP RUN AND FAULT SIGNALS TO THE FIRE ALARM PANEL.



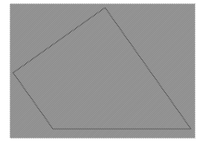
CIVIL DEFENCE

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GENERAL NOTES

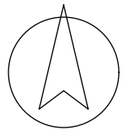
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KEY PLAN



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REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE WATER SCHEMATIC

PUMP SET AND RISER DIAGRAM

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
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DATE
18/10/2026

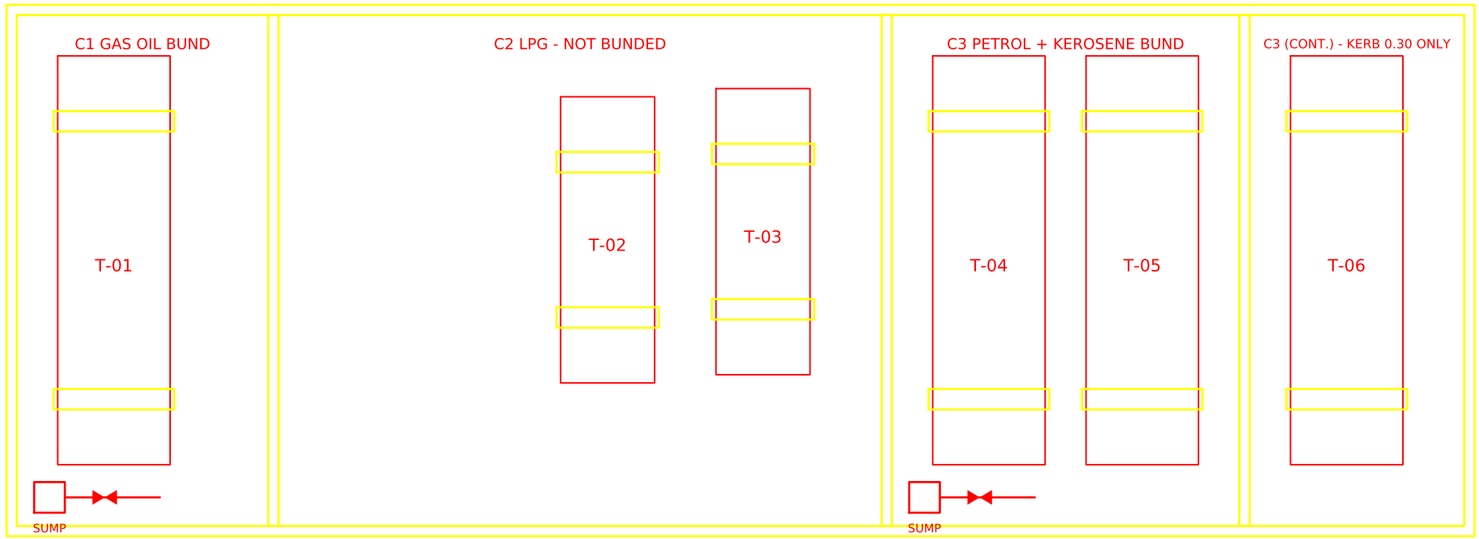
PHASE
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STATUS
FOR REVIEW

DRAWING No.
F-006

SHEET
06
OF 8

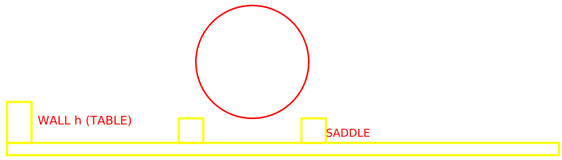
TANK AREA - BUND PLAN 1:150



BUND	TANKS	AREA m2	LARGEST x 1.10	h MIN	WALL h	NET m3
C1	T-01	76.9	59.4	0.82	1.00	70.6
C3	T-04 T-05 T-06	175.0	59.4	0.35	0.60	102.3

1. NET CAPACITY = FLOOR AREA x h - TANK VOLUME BELOW h (TANK BOTTOM 0.60 m ABOVE FLOOR) - SADDLES.
2. REQUIRED: 110 % OF THE LARGEST TANK (NFPA 30 22.11.2 ASKS 100 %); WALL h = h MIN + 0.15 FREEBOARD.
3. C3a AND C3b FORM ONE BUND; THE WALL BETWEEN THEM IS CUT DOWN TO A 0.30 m KERB.
4. LPG COMPARTMENT C2 IS NOT BUNDED: FLOOR FALLS 1 % AWAY FROM UNDER THE TANKS TO AN OPEN GULLY OUTSIDE (NFPA 58 6.4.5 - NO LPG CONTAINER INSIDE A FLAMMABLE LIQUID DIKE).
5. BUND FLOOR 1 % TO THE SUMP; WALLS AND FLOOR LIQUID-TIGHT RC TO ACI 350 (S-005).
6. DRAIN VALVES KEPT CLOSED AND LOCKED; RAINWATER RELEASED ONLY AFTER INSPECTION.

TYPICAL BUND SECTION 1:150



0 1.5 3 4.5 6 7.5m

SCALE 1:150 (A3)

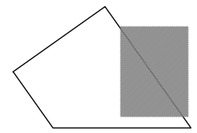
CIVIL DEFENCE

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GENERAL NOTES

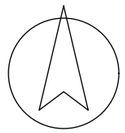
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LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

BUND COMPARTMENTS

CAPACITY CHECK, DRAIN VALVES, SUMPS

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE 1:150

DATE 18/10/2026

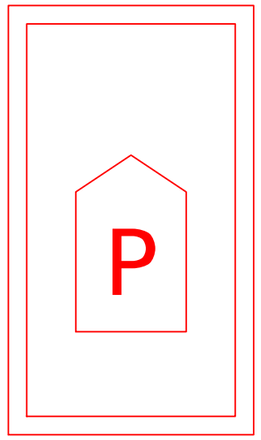
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STATUS FOR REVIEW

DRAWING No. F-007

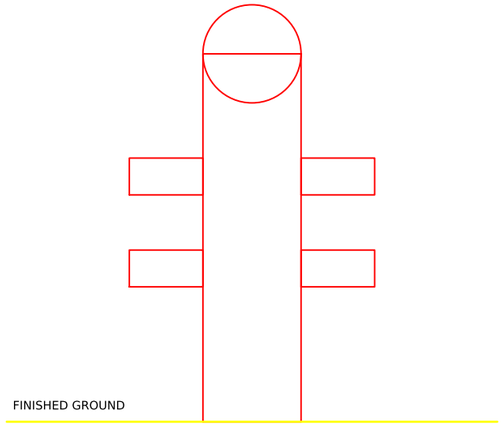
SHEET 07 OF 8

1 EXTINGUISHER CABINET



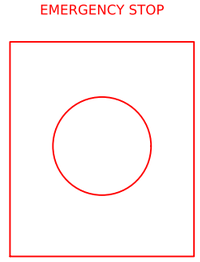
STEEL CABINET, RED, 1.2 mm
GLASS FRONT, TOP <= 1.5 m ABOVE GROUND
PLATE: 'EXTINGUISHER - طفاية حريق'

2 PILLAR HYDRANT



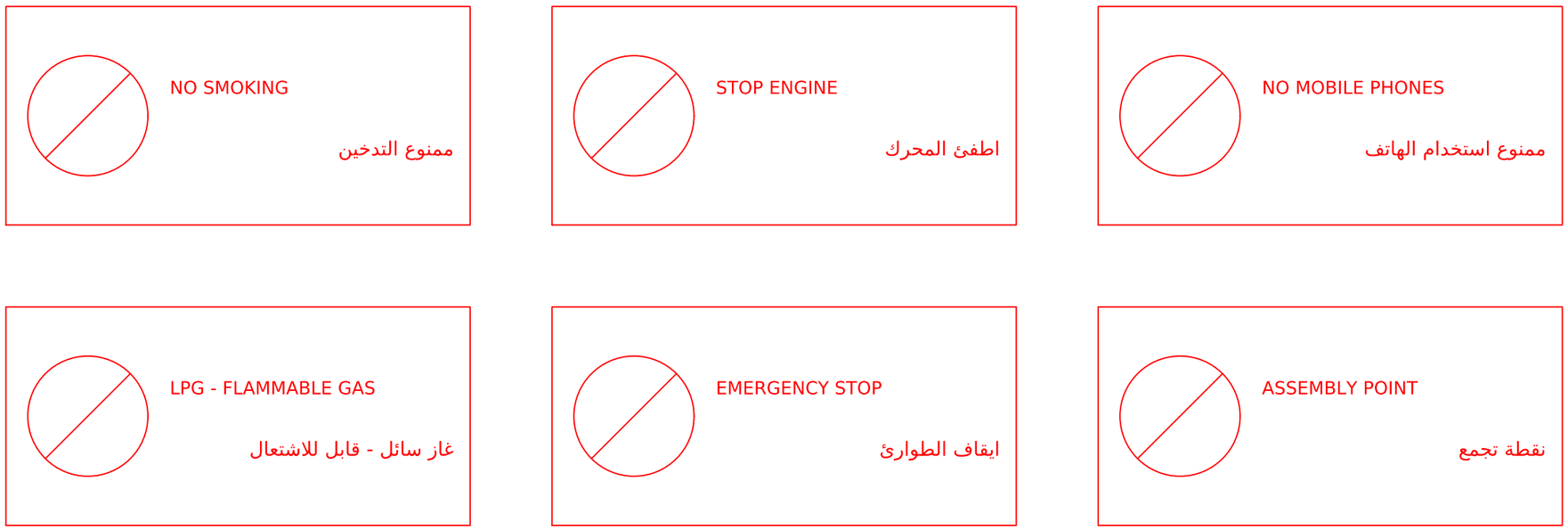
2 x 65 mm OUTLETS, BREAKABLE BASE, DRAIN VALVE; ISOLATING VALVE IN A CHAMBER 1.5 m AWAY; THRUST BLOCK; 0.9 m MIN COVER.

3 ESD PUSH BUTTON



Ex d IIB T4 STATION, RED MUSHROOM, LATCHING ON A POST 1.2 m ABOVE GROUND; ALL BUTTONS IN SERIES IN THE DISPENSER CONTROL CIRCUIT.

4 SAFETY SIGNS (MIN. 600 x 400 mm, REFLECTIVE)



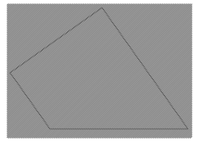
CIVIL DEFENCE

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GENERAL NOTES

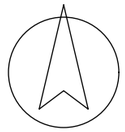
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PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
CIVIL DEFENCE DETAILS
SIGNS, CABINETS, MOUNTING

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
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